

BMU–GW BMS gateway

Rev B0 | 2026–10–04 | Schematics authored by Chinmoy Bhuyan | Isolated CAN FD + RS–485 between CTRL–C2000 and the battery BMS

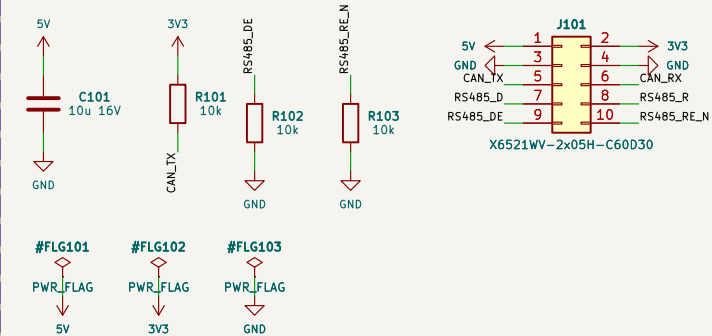
01 Isolated CAN + RS–485

Host header, push–pull isolated 5 V, ISO1042 CAN FD, ISO1410 RS–485, TVS, terminations, field connector

File: sheets/01_gateway.kicad_sch

The field side (GND_BMS) floats with the BMS communication ground; only U–isolators and the bias transformer cross the barrier (checked on every build).
The isolated 5 V is unregulated (5 V x 1.1 minus one Schottky drop): 4.8–5.5 V over a 5 V +/-5 % input, inside the 4.5–5.5 V VCC2 window of ISO1042. R preload keeps it bounded at no load.

Not bench–validated. PELV = controller side, BMS = field side. Sheet 1 Schematics authored by Chinmoy Bhuyan Barrier: 5 kVrms reinforced isolators; bias transformer 2500 VAC test (functional/basic)		
Sheet: / File: BMU–GW.kicad_sch		
Title: BMU–GW BMS gateway – Cover / Hierarchy		
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The schematic diagram illustrates the CAN module's internal components and their connections. The U102 (ISO1042DWR) transceiver is connected to a 3V3 supply, CAN_TX, CAN_RX, and GND. It has four 100nF 50V capacitors (C106, C107, C108, C109) connected to its VCC1, VCC2, CANH, and CANL pins respectively. The D103 (ESD2CAN24BZRQ1) diode is connected to the CAN_H and CAN_L lines. The JP101 solder jumper is shown open by default. The R105 and R106 resistors are 60.4R.

The schematic diagram illustrates the internal components and connections of the ISO1410DWR module. It features two main integrated circuits: U103 (ISO1410DWR) and D104 (CDSOT23-SM712).

U103 (ISO1410DWR) Connections:

- VCC1 (Pin 1):** Connected to 3V3.
- D (Pin 6):** Connected to RS485_D.
- DE (Pin 5):** Connected to RS485_DE.
- RE (Pin 4):** Connected to RS485_RE_N.
- RS485_R (Pin 3):** Connected to RS485_R.
- NC (Pin 7):** Not connected (marked with an 'X').
- GND1 (Pin 2):** Connected to GND.
- GND1 (Pin 8):** Connected to GND.
- VCC2 (Pin 16):** Connected to 5V_BMS.
- A (Pin 12):** Connected to RS485_A.
- B (Pin 13):** Connected to RS485_B.
- NC (Pin 10):** Not connected (marked with an 'X').
- NC (Pin 11):** Not connected (marked with an 'X').
- NC (Pin 14):** Not connected (marked with an 'X').
- GND2 (Pin 15):** Connected to GND_BMS.
- GND2 (Pin 9):** Connected to GND_BMS.

D104 (CDSOT23-SM712) Connections:

- I01 (Pin 1):** Connected to RS485_A.
- I02 (Pin 2):** Connected to RS485_B.
- COM (Pin 3):** Connected to GND_BMS.

Other Components and Connections:

- 3V3:** Power supply connection.
- 5V_BMS:** BMS power supply connection.
- GND:** Ground connection.
- GND_BMS:** BMS ground connection.
- C110:** 100nF 50V capacitor connected between 3V3 and GND.
- C111:** 100nF 50V capacitor connected between 5V_BMS and GND_BMS.
- JP102:** Solder jumper, open by default, connected between RS485_A and RS485_TA.
- R107:** 120R resistor connected between RS485_TA and RS485_B.

The schematic diagram illustrates the CAN bus termination and filtering circuit. It shows two CAN bus lines, CAN_H and CAN_L, each terminated with a 1MΩ resistor (R108) to GND_BMS. A 4.7nF capacitor (C112) is connected between the two lines. A J102 connector is shown with pins 1 to 6 connected to CAN_H, CAN_L, GND_BMS, RS485_A, RS485_B, and SHIELD respectively. The RS485_A and RS485_B pins are connected to a common ground point labeled KF2EDGR-3.81-6P.

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